Examiner only

4. There are 33 known isotopes of krypton. One of the isotopes – ^{81}Kr – decays by electron capture.

(a) Write an equation to show this decay. [1]

....

(b) Krypton-81 has been used for dating old groundwater. It takes 6.87×10^5 years for 2.0 g of ^{81}Kr to decay to 0.25 g. Calculate its half-life. [1]

Half-life = ... years

5. Boron has a relative atomic mass of 10.8. It has only two naturally-occurring isotopes, one of which has an abundance of 80.0% and a mass of 11. Calculate the mass of the second isotope. [2]

Mass = ...

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